

requirements and remain relevant for the next decade.

Hence the option of using private 5G networks to provide connectivity for the motorsports industry cars is an attractive proposition. 5G is an option for the motorsports industry as it is designed to overcome and mitigate some of the issues involved in using wireless communications technologies for the motorsports use cases.

Proposed solution architecture

The figure on previous page shows the proposed ORAN disaggregated architecture for the motorsports use case. The solution uses the 7.2 split architecture, which is modular and extensible for the motorsports use case.

In this architecture, the distributed units (DUs) run some of the physical layer algorithms besides the radio link control (RLC) and media access control (MAC) layers. The DUs are housed in the track side units (TSU), which also provide the power for the DUs.

The centralised unit (CU) runs the radio resource control (RRC) and packet data convergence protocol (PDCP) layers. There is a redundant 25G fibre ring to backhaul the traffic from the DU to the CU. The CU and 5G core are centralised at the event technical centre.

This split architecture makes the solution scalable as the number of radios will depend on the track layout and dimensions. The CU can control multiple DUs and run the required algorithms specific to the motorsports use cases.

Design considerations

Some of the design considerations and proposed solutions when us-

System Bandwidth Calculations

The following table shows the system bandwidth required for a typical racing circuit.

SYSTEM CHARACTERISTICS		
UL/DL	Uplink	Downlink
FDD/TDD	TDD	
Number of component carriers	1	
MIMO number of layers	2	
MU-MIMO beamforming	Yes, 2 beams	
Modulation order	16QAM	
Sub-carrier spacing	60kHz	
Per channel bandwidth	15MHz	
Slot format (uplink heavy)	Format 11	
Coding rate (robust coding)	658/1024 (robust coding)	
Per radio unit throughput	90Mbps	10Mbps
Number of cars per channel	2	
Throughput per car	45Mbps	5Mbps
Number of channels for 28 cars	14	
System bandwidth (standard guard band)	210MHz (14x15MHz)	

If using a higher order 64QAM modulation, the system bandwidth required is approx. 100MHz.

ing 5G to provide the throughput requirements for racing cars are described below.

High speed. The cars move at a very high speed on track, up to 350 kilometres an hour. This high speed introduces unique design challenges in the design of a 5G solution. The Doppler shift or Doppler spread is a concern because of the shift in the frequency of the signals sent and received by the fixed radios from the moving cars. The Doppler shift is directly proportional to the speed of the car. This also results in inter-carrier interference.

Solution. Algorithms need to be designed to compensate for the Doppler shift introduced due to the speed at which the cars are travelling. The Doppler shift depends on the speed and direction of movement of the car relative to the trackside radio. Thus, the Doppler results in an increase or a decrease in the frequency, depending on the direction and speed of movement of the car relative to the radio. The

transmit frequency is adjusted in the RF front end to compensate for the Doppler shift.

Street circuits. Several racing tracks, for example in Formula One, are in cities where there is infrastructure like buildings, fences, trees, grandstands, and street furniture. This infrastructure causes interference and scattering of the 5G radio signals to and from the cars, degrading the signal quality.

Solution. Antenna diversity and multipath reception needs to be implemented to maintain the quality of the received signals at the radio and the UE. The channel state information (CSI) can also be computed and predicted more frequently to determine the channel

condition. Machine language algorithms can be used to make a more accurate estimation of the channel condition. This will require a higher compute power at the base station to accurately calculate the CSI and compensate accordingly.

Handover. A unique challenge faced due to the high speed at which the car travels is that the time for handover between the neighbouring cells is small. It is of the order of a few tens of milliseconds to prevent drops or glitches in the telemetry audio and video data that is sent from the car. Hence the seamless handover needs to be implemented for these racing cars.

Solution. Multi-TRxP features like co-ordinated multi-point (CoMP) and multi-panel beamforming can be used to cancel out the interference from neighbouring cars. CoMP utilises different techniques to dynamically coordinate the transmission and reception for a particular UE to and from multiple base stations.